ENV S 193DS Final

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https://github.com/stellassymonds/ENVS-193DS_spring-2026_final

```
# read in packages
library(tidyverse) # general use
library(janitor) # cleaning data frames
library(here) # file/folder organization
library(DHARMa) # for residual plots
library(ggeffects) # getting model predictions
library(MuMIn) # model selection
library(effsize) # effect size

# read in data
nests <- read.csv(here("data", "nest_data_final.csv"))
bike_rides <- read.csv(here("data", "bike_rides.csv"))
```

Problem 1

a. In part 1, they used a generalized linear model. In part 2, they used a chi-squared test.

b. Part 1: We found evidence to suggest that the distance to water edge does predict probability of American Avocet microhabitat use.

Part 2: We found evidence to suggest that there is a difference between American Avocet, Forster's Tern, and Black-necked Stilt bird species' habitat preferences of managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat habitats.

c. Associated behavior type of either feeding, breeding, or roosting could be a categorical variable that influences the probability of American Avocet microhabitat use. For example, waterbirds that are breeding or roosting may prefer islands wihtin ponds.

Problem 2

a.

```
# create new object from nests
nests_clean <- nests |>
  # reordering ant species
  mutate(ant_species = as_factor(ant_species),
         ant_species = fct_relevel(ant_species,
                                   "AB", # C. sjostedti
                                   "RRB", # C. mimosae
                                   "BBR")) |> # C. nigriceps
  # select for preferred columns
  select(height_cm, ant_species, case_control) |>
  # filter for preferred ant species
  filter(ant_species %in% c("RRB", "BBR", "AB")) |>
  # create new column with full scientific name
  mutate(scientific_name = case_when(
    ant_species == "AB" ~ "C. sjostedti",
    ant_species == "RRB" ~ "C. mimosae",
    ant_species == "BBR" ~ "C. nigriceps"))

slice_sample(nests_clean, # data frame to display
             n = 10) # select ten random rows
```

```
   height_cm ant_species case_control scientific_name
1      183.0         RRB            0      C. mimosae
2      212.5         RRB            1      C. mimosae
3      316.0         RRB            0      C. mimosae
4      196.5         RRB            0      C. mimosae
5      456.0          AB            0    C. sjostedti
6       54.0         BBR            0    C. nigriceps
7      173.5         RRB            0      C. mimosae
8      223.5         RRB            0      C. mimosae
9      368.0         RRB            0      C. mimosae
10     109.0         RRB            0      C. mimosae
```

b. 1s mean that the tree contained a nest and 0s mean the tree did not contain a nest and was available for comparison.

c. Tree height is a continuous variable. As tree height increases, the probability of nest occurrence would increase because they are farther from possible nest predators or disturbances (Results, paragraph 2).

Acacia ant species is a categorical variable. If an ant species is a more aggressive defender, the probability of nest occurrence would increase because the ant species reduces the risk of nest predation (Introduction, paragraph 3). If acacia ants defend against all possible tree disturbances, then the probability of nest occurrence in trees with less aggressive ant species would increase (Introduction, paragraph 3).

d. 2 models total:

Model number	Tree height	Ant Species	Interaction?
1	X	X	no
2	X	X	yes

e.

f.

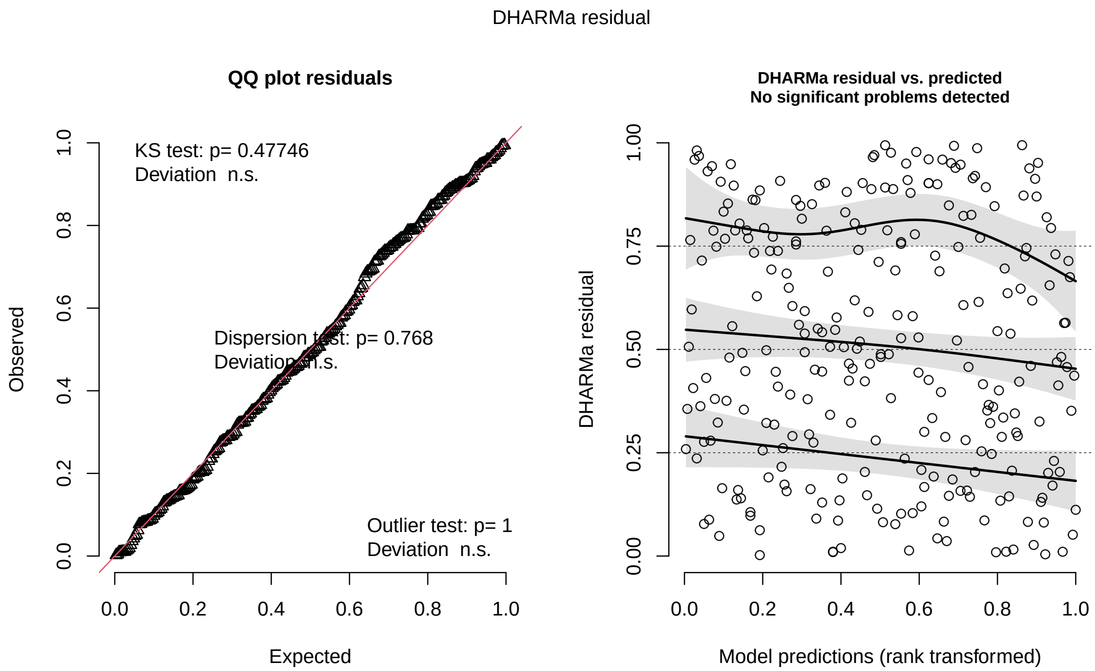
```
AICc( # calculating AIC of both models
  model1,
  model2) |>
  # arranging output in order of AIC
  arrange(AICc)
```

```
       df     AICc
model2  6 238.1236
model1  4 258.8940
```

The best model as determined by Akaike's Information Criterion (AIC) shows nest occurrence as a function of tree height in cm interacting with ant species.

g.

```
# generating diagnostic plots for best model (by AIC value) from DHARMa
plot(
  simulateResiduals(model2)
  )
```



h.

i. Figure 1. Nest occurrence as a function of tree height in cm interacting with ant species. Describe. Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Data, code, and metadata for: Symbiotic acacia ants drive nesting behavior by birds in an African savanna [Data set]. Zenodo. https://doi.org/10.5281/zenodo.8373322

j.

k.

Problem 3

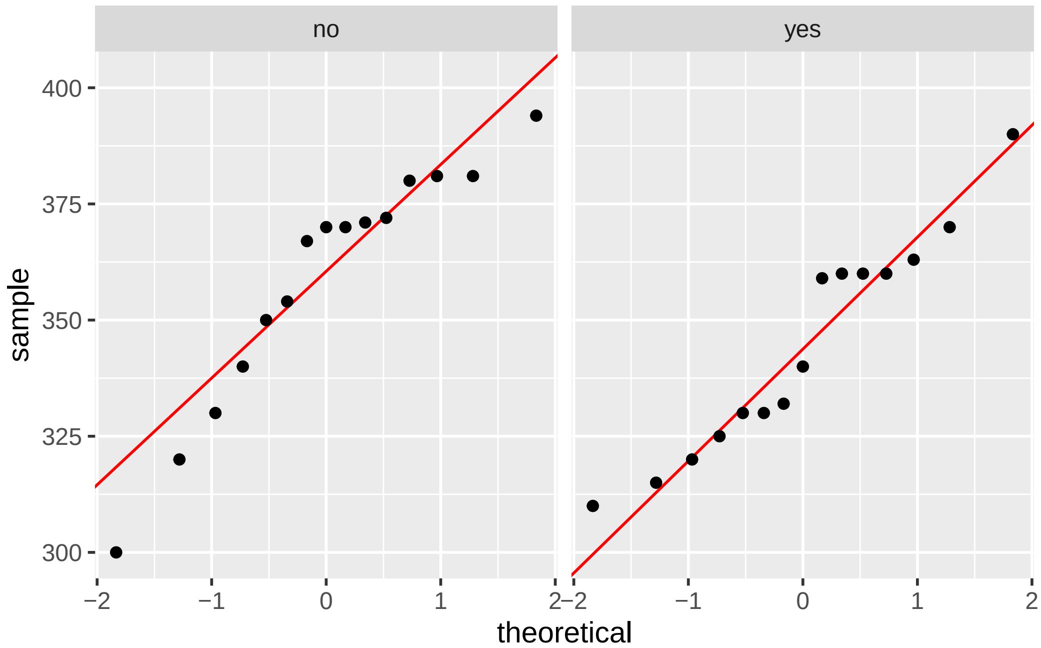
a. My affective visualization would be impossible to analyze without a key accompanying it. I prioritized creativity over feasibility of understanding. My exploratory visualizations are very easy to understand and make conclusions (or lack thereof) from. All my visualizations showed close to an equal amount of bike rides when I did and didn't have plans after class and low, medium, or high wind speed. The variation in data was consistent and can be seen in all visualizations. A slight difference in medians between bike rides when I have plans after class and no plans after class is visible in my exploratory visualization, but is not clear in my affective visualization. This is because my data is oriented around a central point in my affective visualization. I don't think a clear association between wind speed and bike ride duration is obvious in any of my visualizations. My feedback in workshop consisted of possibly adding icons to the end of the lines to represent temperature or another variable. I ultimately decided to not implement this feedback, since neither temperature nor other variables I tracked had any visible affect on my data, and I thought it may make it harder to see the difference between the lengths of each stitched line.

b. A two sample t-test is appropriate to test whether there is a difference in mean bike ride duration between two groups, categorized by status of my plans after class (yes/no). My plans status after class is a binary variable, meaning it can be split into two groups, and my bike ride duration is a continuous variable.

c.

```
# cleaning data
bike_rides_clean <- bike_rides |> # create new object from rides
  clean_names() |> # clean column names
  # select plans and duration columns
  select(plans_after_class, duration_seconds)

# normality check
# base layer: ggplot
ggplot(data = bike_rides_clean, # starting data frame
       aes(sample = duration_seconds)) + # y-axis for QQ plot
  # first layer: QQ reference line
  geom_qq_line(color = "red") + # adding a color
  geom_qq() + # second layer: QQ plot
  # creating "facets"
  facet_wrap(~ plans_after_class) # show treatments separately
```




```
# check variances with F test of equal variances
var.test(
  duration_seconds ~ plans_after_class, # formula: response variable ~ grouping variable
  data = bike_rides_clean) # bike_rides_clean data frame
```

```

    F test to compare two variances

data:  duration_seconds by plans_after_class
F = 1.253, num df = 14, denom df = 14, p-value = 0.6789
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.4206592 3.7320819
sample estimates:
ratio of variances
           1.25297
```

```
t.test(
  # formula: response variable ~ grouping variable
  duration_seconds ~ plans_after_class,
  var.equal = TRUE, # argument for unequal variances
  data = bike_rides_clean)  # bike_rides_clean data frame
```

```
    Two Sample t-test

data:  duration_seconds by plans_after_class
t = 1.5915, df = 28, p-value = 0.1227
alternative hypothesis: true difference in means between group no and group yes is not equal
95 percent confidence interval:
 -4.134173 32.934173
sample estimates:
 mean in group no mean in group yes
         358.6667          344.2667
```

```r
# calculate cohen's d
d <- cohen.d(duration_seconds ~ plans_after_class, data = bike_rides_clean)

d # display effect size value
```

```

Cohen's d

d estimate: 0.5811322 (medium)
95 percent confidence interval:
     lower      upper
-0.1824647  1.3447291
```

d.

```r
# create new object from bike_rides_clean
bike_rides_clean_summary <- bike_rides_clean |>
  group_by(plans_after_class) |>
  summarize( # what I want in the summary
    mean = mean(duration_seconds),
    n = length(duration_seconds),
    sd = sd(duration_seconds)) |>
  ungroup() # ungroup the data frame

# base layer: ggplot
ggplot()
```

e. Figure 2. Bike ride duration dependent on whether I had plans after class.

f. (two sample t-test, t()=critical, $/alpha = 0.05$)